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DSC630-T301 Predictive Analytics
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Code from Pycharm

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# Megan Pokal, DSC 630, Assignment 1
# California Housing Prices Data Project

# Imported Libraries needed
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import zipfile

# Direct Zip Pathway on my Computer
# I got this dataset from Kaggle: https://www.kaggle.com/datasets/fedesoriano/california-
housing-prices-data-extra-features?resource=download
zip_path = "/Users/meganpokal/PyCharmMiscProject/.venv/California_Houses.csv"


# Pull CSV file directly
with zipfile.ZipFile(zip_path) as z:
    with (z.open("California_Houses.csv") as f):
        df = pd.read_csv(f)

# Preview dataset
print(df.head())

# Pull info about dataset
df.info()

# Print list of columns
print(list(df.columns))

# Statistics Summary
print(df.describe(include="all"))

# Histogram: Looking at Median House Value
plt.figure(figsize=(8,6))
plt.hist(df["Median_House_Value"], bins=50, edgecolor="black")
plt.title("Distribution of Median House Value")
plt.xlabel("House Value")
plt.ylabel("Number of Houses")
plt.show()
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# Boxplot: Looking at Distance to from the Coast vs House Value
# I had to take 2000 random rows to avoid overplotting
sample_df = df.sample(2000, random_state=42)

plt.figure(figsize=(10,6))
sns.boxplot(x=pd.cut(sample_df["Distance_to_coast"], bins=10),
        y=sample_df["Median_House_Value"])
plt.title("House Values by Distance to Coast")
plt.xlabel("Distance to Coast")
plt.ylabel("Median House Value")
plt.xticks(rotation=45)
plt.show()

# Bivariate Plot (Scatterplot): Looking at Income vs House Value
plt.figure(figsize=(10,6))
plt.scatter(sample_df["Median_Income"], sample_df["Median_House_Value"],
        alpha=0.4, c="teal")
plt.title("Income vs House Value")
plt.xlabel("Median Income")
plt.ylabel("Median House Value")
plt.show()

# Heatmap (Additional Graph): Looking at 7 Columns to not overplot
numeric_cols = ["Median_House_Value", "Median_Income", "Median_Age",
           "Tot_Rooms", "Tot_Bedrooms", "Population", "Households"]
plt.figure(figsize=(8,6))
sns.heatmap(df[numeric_cols].corr(), annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Heatmap of Key Housing Features")
plt.show()
```

Output from Code:

```
   Median_House_Value  Median_Income  Median_Age  Tot_Rooms  Tot_Bedrooms  \
0            452600.0         8.3252          41        880           129
1            358500.0         8.3014          21       7099          1106
2            352100.0         7.2574          52       1467           190
3            341300.0         5.6431          52       1274           235
4            342200.0         3.8462          52       1627           280

   Population  Households  Latitude  Longitude  Distance_to_coast  \
0         322         126     37.88    -122.23        9263.040773
1        2401        1138     37.86    -122.22       10225.733072
2         496         177     37.85    -122.24        8259.085109
3         558         219     37.85    -122.25        7768.086571
4         565         259     37.85    -122.25        7768.086571

   Distance_to_LA  Distance_to_SanDiego  Distance_to_SanJose  \
0   556529.158342         735501.806984         67432.517001
1   554279.850069         733236.884360         65049.908574
2   554610.717069         733525.682937         64867.289833
3   555194.266086         734095.290744         65287.138412
4   555194.266086         734095.290744         65287.138412

   Distance_to_SanFrancisco
0              21250.213767
1              20880.600400
2              18811.487450
3              18031.047568
4              18031.047568
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20640 entries, 0 to 20639
Data columns (total 14 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Median_House_Value        20640 non-null  float64
 1   Median_Income             20640 non-null  float64
 2   Median_Age                20640 non-null  int64
 3   Tot_Rooms                 20640 non-null  int64
 4   Tot_Bedrooms              20640 non-null  int64
 5   Population                20640 non-null  int64
 6   Households                20640 non-null  int64
 7   Latitude                  20640 non-null  float64
 8   Longitude                 20640 non-null  float64
 9   Distance_to_coast         20640 non-null  float64
 10  Distance_to_LA            20640 non-null  float64
 11  Distance_to_SanDiego      20640 non-null  float64
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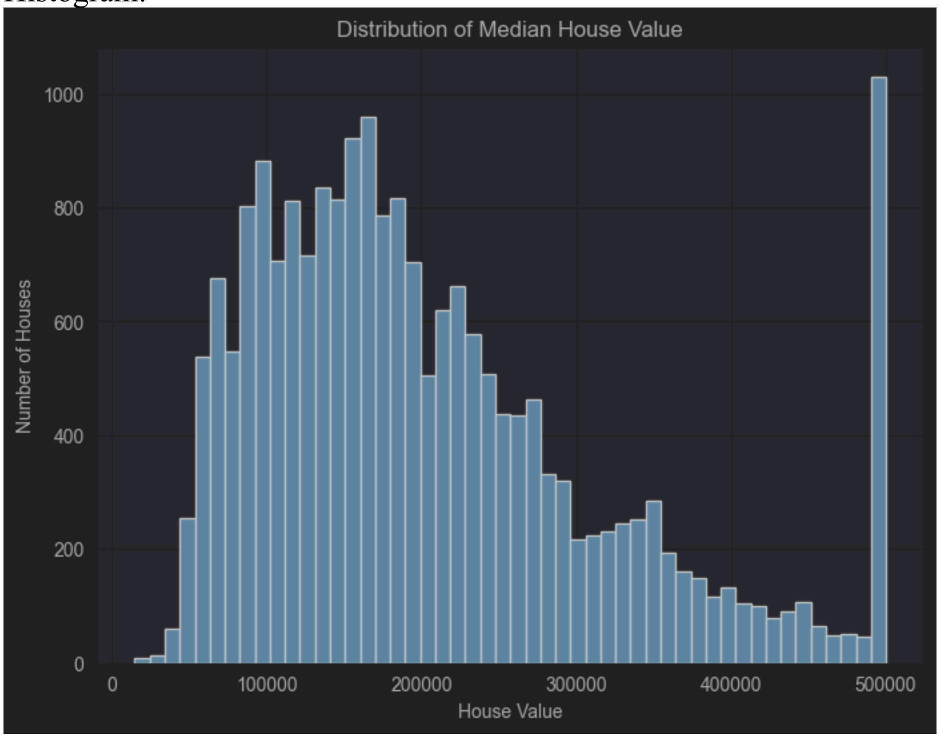
 12  Distance_to_SanJose       20640 non-null  float64
 13  Distance_to_SanFrancisco  20640 non-null  float64
dtypes: float64(9), int64(5)
memory usage: 2.2 MB
['Median_House_Value', 'Median_Income', 'Median_Age', 'Tot_Rooms', 'Tot_Bedrooms',
'Population', 'Households', 'Latitude', 'Longitude', 'Distance_to_coast', 'Distance_to_LA',
'Distance_to_SanDiego', 'Distance_to_SanJose', 'Distance_to_SanFrancisco']
       Median_House_Value  Median_Income    Median_Age     Tot_Rooms  \
count        20640.000000   20640.000000  20640.000000  20640.000000
mean        206855.816909       3.870671     28.639486   2635.763081
std         115395.615874       1.899822     12.585558   2181.615252
min          14999.000000       0.499900      1.000000      2.000000
25%         119600.000000       2.563400     18.000000   1447.750000
50%         179700.000000       3.534800     29.000000   2127.000000
75%         264725.000000       4.743250     37.000000   3148.000000
max         500001.000000      15.000100     52.000000  39320.000000

       Tot_Bedrooms    Population    Households      Latitude     Longitude  \
count  20640.000000  20640.000000  20640.000000  20640.000000  20640.000000
mean     537.898014   1425.476744    499.539680     35.631861   -119.569704
std      421.247906   1132.462122    382.329753      2.135952      2.003532
min        1.000000      3.000000      1.000000     32.540000   -124.350000
25%      295.000000    787.000000    280.000000     33.930000   -121.800000
50%      435.000000   1166.000000    409.000000     34.260000   -118.490000
75%      647.000000   1725.000000    605.000000     37.710000   -118.010000
max     6445.000000  35682.000000   6082.000000     41.950000   -114.310000

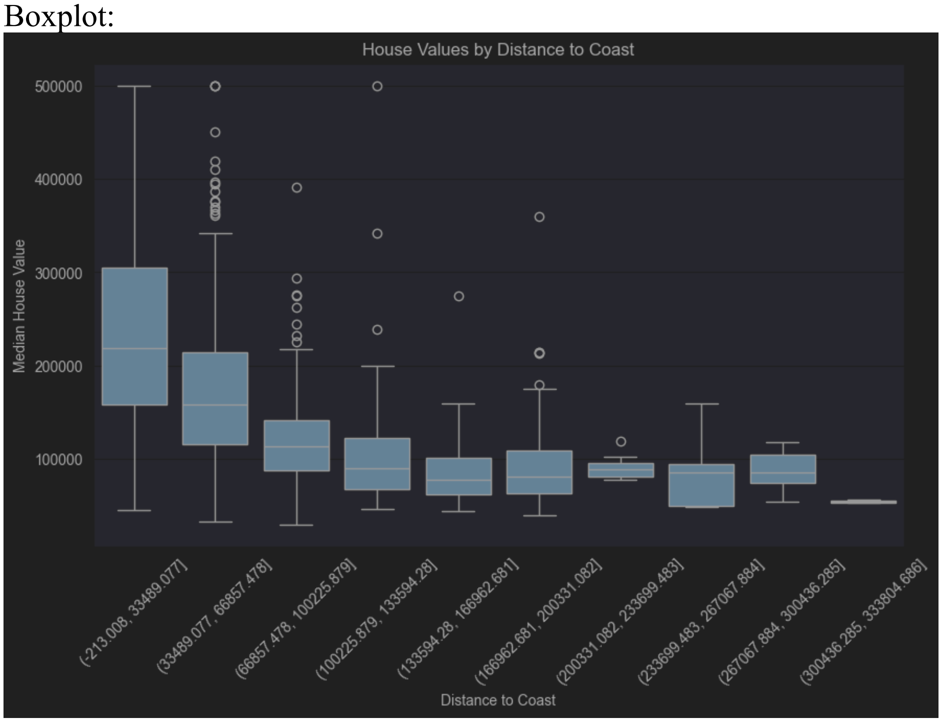
       Distance_to_coast  Distance_to_LA  Distance_to_SanDiego  \
count       20640.000000    2.064000e+04          2.064000e+04
mean        40509.264883    2.694220e+05          3.981649e+05
std         49140.039160    2.477324e+05          2.894006e+05
min           120.676447    4.205891e+02          4.849180e+02
25%          9079.756762    3.211125e+04          1.594264e+05
50%         20522.019101    1.736675e+05          2.147398e+05
75%         49830.414479    5.271562e+05          7.057954e+05
max        333804.686371    1.018260e+06          1.196919e+06

       Distance_to_SanJose  Distance_to_SanFrancisco
count         20640.000000              20640.000000
mean         349187.551219             386688.422291
std          217149.875026             250122.192316
min             569.448118                456.141313
25%          113119.928682             117395.477505
50%          459758.877000             526546.661701
75%          516946.490963             584552.007907
max          836762.678210             903627.663298
```

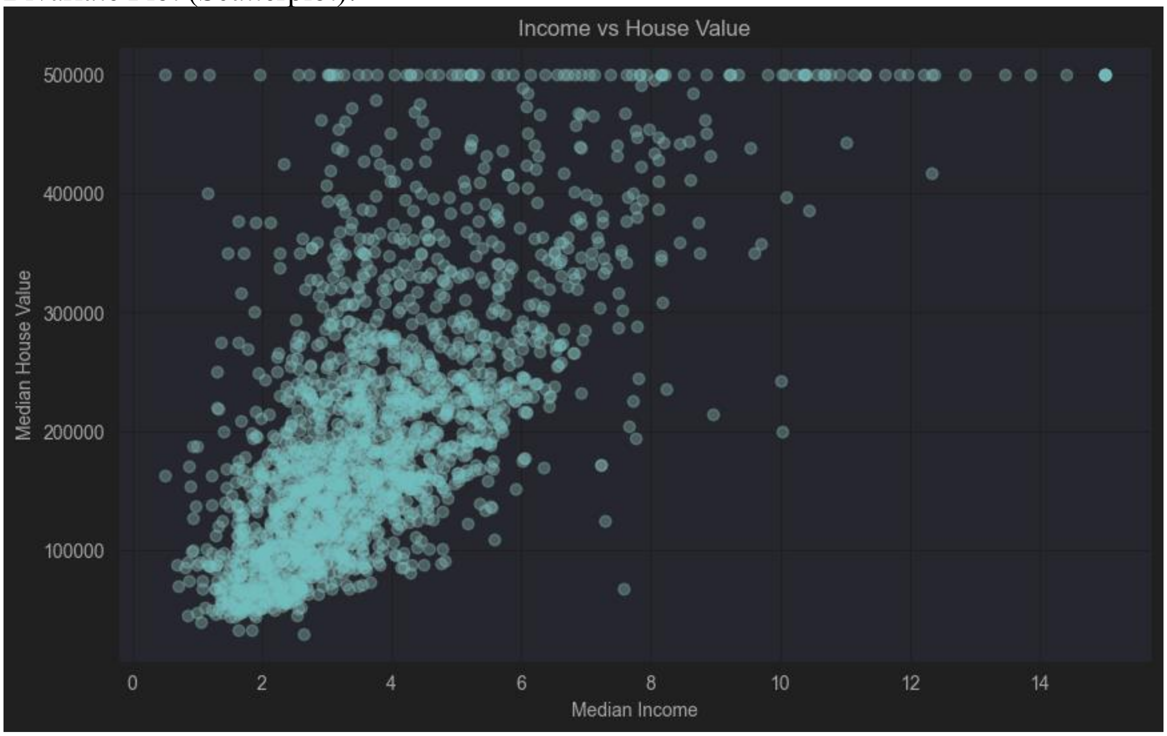
Histogram:



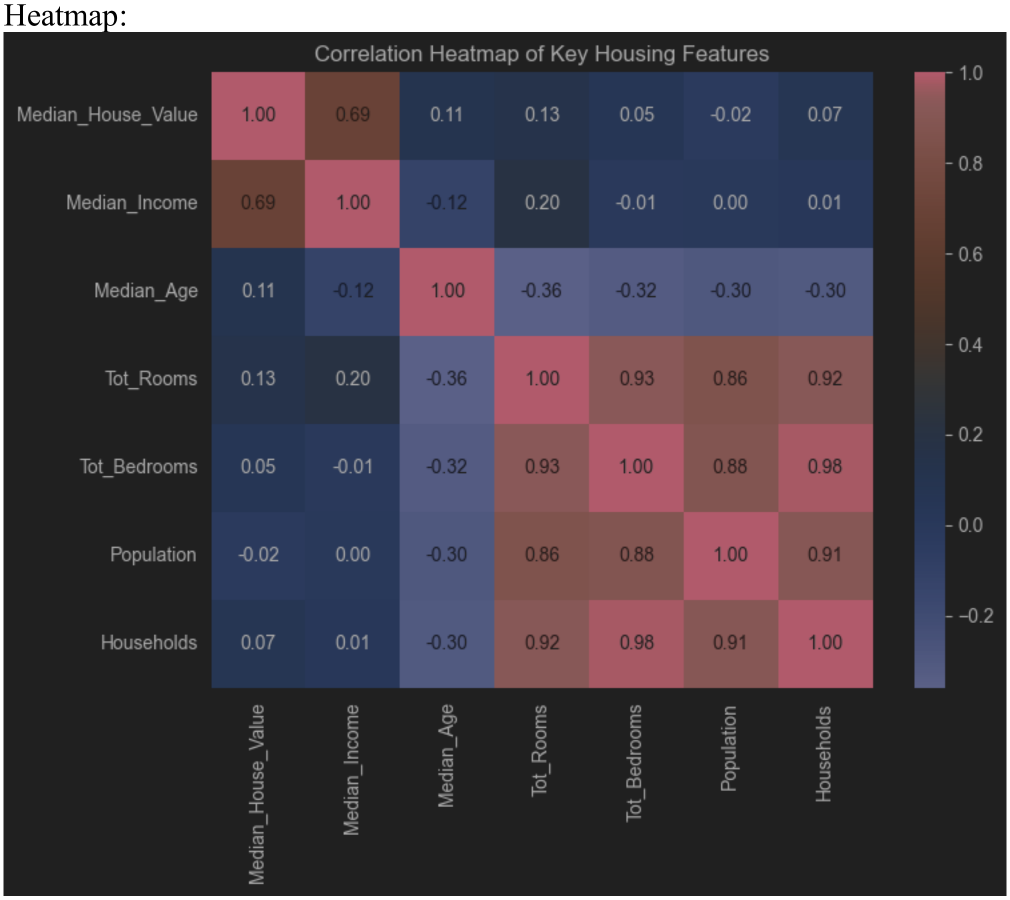
Boxplot:



Bivariate Plot (Scatterplot):



Heatmap:



Conclusion from Results:

Reason on why I choose this data, I love San Diego and Real Estate. Little fun side note, I live in La Jolla, California which is about a 10 to 15-minute drive to the beach depending on which one I want to go to and how fast I drive… I drive a 2024 Mustang Mache, Electric, Bright Blue, so SHE moves. So, I contest to this data / finding’s below.

1. Histogram Graph
    a. This graph was showing a distribution of the Number of Houses and House Values. The range of House Values was between $100,000 to $500,000 but most house fell between $100,000 and about $300,000 range.
    b. This means that the graph is a Right Skewed Histogram, which is a positively skewed histogram.
2. Boxplot Graph
    a. This graph was showing the Median House Values and the Distance from the Coast in a Boxplot. From the dataset, Distance_to_Coast had too many values in the data, so I made the decision to only look at 2,000 random values from the data.
    b. From looking at the boxplot, it shows the further you get from the Coast the Median value of the House starts to drop with it. Which makes sense, the further you have to drive to the beach, the cheaper the house.
3. Bivariate Plot (Scatterplot) Graph
    a. This graph is showing the Median Income and the House Value.
    b. The Graph is showing a strong positive relationship between Median Income and the House Value, so a household income is a strong factor that the houses will be more expenses in that area.
4. Heatmap Graph
    a. This Heatmap is my additional graph and even though I a good data set where I didn’t have to clean it, this Heatmap gave me the most headache. Until I figured out, I needed to code it a little differently and use a different column, so I picked 7 at random.
    b. It shows on the map that correlation between Median House Value is Most Strongly Correlated with Median Income with a score of 0.69 on the Heatmap, in the brown square.
    c. The heatmap is showing the most related features from the dataset that are related to the Housing Prices. It confirms that income is a key value when it comes to house values.